Haonan Li
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CURRICULUM VITAE

Haonan Li, PhD

PROFESSIONAL DATA

Johns Hopkins University
Bloomberg School of Public Health
Department of Environmental Health & Engineering
615 N. Wolfe Street, Baltimore, Maryland 21205
hli263@jh.edu

RESEARCH INTERESTS

Integrating multi-omics approaches (metabolomics, microbiome, and bioactive molecules) with longitudinal environmental epidemiology to elucidate biological mechanisms linking environmental exposures, including air pollution and the chemical exposome, to cardiometabolic and neurodevelopmental health

EDUCATION AND TRAINING

Degree	***Year***	***Institution***	***Field***
PhD	2023	Peking University	Environmental Epidemiology, Bioactive Lipidomics, Air Pollution
BS	2018	Xiamen University	Marine Biology, Environmental Microbiology

Postdoctoral Training

Years	***Department***	***Institution***
2024 – Present	Environmental Health and Engineering	Johns Hopkins University

PROFESSIONAL ACTIVITIES

SOCIETY MEMBERSHIP

2026 – Present	Member, International Society of Exposure Science (ISES)
2025 – Present	Member, International Society for Environmental Epidemiology (ISEE)
2025 – Present	Member, American Society for Mass Spectrometry (ASMS)

JOHNS HOPKINS MEMBERSHIP

2026 – Present	NIEHS P30 Center Member, Community Health Addressing Regional Maryland Environmental Determinants of Disease (CHARMED)

EDITORIAL ACTIVITIES

JOURNAL PEER-REVIEW ACTIVITIES

Environmental Health Perspectives
Journal of Hazardous Materials
Journal of Exposure Science & Environmental Epidemiology
Journal of Child Psychology and Psychiatry
Frontiers in Public Health

PROPOSAL REVIEWS

Children's Hospital Los Angeles, Los Angeles, CA

2025 Grant Reviewer, Pilot Project Program of Southern California Center for Latino Health

HONORS AND AWARD

2022 Award for Scientific Research (Top 10%)
Peking University

2018 Outstanding Graduate Award (Top 10%)
Xiamen University

2016 National Scholarship (Top ~0.2%)
Ministry of Education of China

PUBLICATIONS

Number of publications: 27; **Number of Citations (Google Scholar index):** 385

H-index (Google Scholar index): 12; **I10-index (Google Scholar index):** 15

JOURNAL ARTICLES (PEER-REVIEWED)

1. **Li H**, Chen X, Xu Y, Wang T, Yao Y, Chen W, Gong J, Qiu X, Zhu T, 2026. Association between ambient temperature and thrombosis-related signaling lipids: Implications for early cardiovascular risk. *Journal of Exposure Science & Environmental Epidemiology*. https://doi.org/10.1038/s41370-026-00987-z. (IF = 5.9)
2. Dong Q, **Li H**, Xu Y, Zhang Y, Gao Y, Wu Z, Ye C, Gong J, Zhu T, 2026. High-altitude exposure: Effects on cognitive performance, neurovascular mechanisms, and mitigation strategies. *Environment & Health*. https://doi.org/10.1021/envhealth.6c00248. (IF = 8.8)
3. **Li H**, Schenker RB, Marquess KM, Paudel D, Ramirez CB, Lurmann F, Ren M, Jang C, Allayee H, Liao J, Chen Z, Goran MI, Alderete TL, 2026. Ambient fine particulate matter components and liver fat and stiffness in youth: Metabolic vulnerability in PNPLA3 risk carriers. *Environment International*, 215: 110438. https://doi.org/10.1016/j.envint.2026.110438. (IF = 10.2)
4. Ottino-González J, Adise S, Descarpentrie A, Esaian S, Marquess KM, Paudel D, **Li H**, Alderete TL, Bansal R, Peterson BS, Goran MI, 2026. Early-life cumulative intake of ultra-processed foods and subcortical brain volume at age 6 y: A prospective cohort study. *American Journal of Clinical Nutrition*, 124(2): 101350. https://doi.org/10.1016/j.ajcnut.2026.101350. (IF = 6.9)
5. **Li H**, Holzhausen EA, Paudel D, Young ND, Ottino-González J, Lurmann FW, Walker DI, Chang HH, Liang D, Bansal R, Goran MI, Peterson BS, Alderete TL, 2026. Associations of prenatal fine particulate matter mixtures with neurodevelopmental outcomes in early childhood: Component- and source-specific insights. *Particle and Fibre Toxicology*, 23: 14. https://doi.org/10.1186/s12989-025-00657-2. (IF = 8.2)
6. Qiao R, Hua Q, Li J, Shi Y, Chen W, Chai Q, Fan Y, Li A, **Li H**, Meng X, Sheng M, Xu R, Xu Y, Yao Y, Zhang Y, Zhang Y, Danzengdunzhu, Zhuoga, Weschler CJ, Zhang J, Shang J, Qiu X, Zhu T, Gong J, Liu Y, 2026. Indoor exposure to ozone-derived carbonyls: Association with indicators of blood and cardiovascular physiology among healthy young adults. *Environmental Science & Technology Air*, 3(2): 517–527. https://doi.org/10.1021/acsestair.5c00369. (IF: not yet indexed)
7. **Li H**, Han Y, Chen W, Wang Y, Xu Y, Wang T, Gong J, Li W, Zhang H, Wang J, Qiu X, Zhu T, 2025. Lysoglycerophospholipid metabolism alterations associated with ambient fine

particulate matter exposure: Insights into the pro-atherosclerotic effects. *Environmental Pollution*, 367: 125646. https://doi.org/10.1016/j.envpol.2025.125646. (IF = 7.3)

8. Xu Y, Han Y, Chen W, **Li H**, Wang Y, Chatzidiakou L, Gong J, Wang T, Zhang H, Yan L, Li Y, Li W, Wu Y, Liu J, Jones RL, Barratt B, Kelly FJ, Zhu T, 2025. Nonoptimal temperature exposure and sphingolipid metabolism: Implications for cardiovascular risk. *Environmental Science & Technology*, 59(41): 21885-21897. https://doi.org/10.1021/acs.est.5c07999. (IF = 11.3)
9. Wang T, Chen X, Yao Y, Chen W, **Li H**, Xu Y, Guan T, Gong J, Qiu X, Zhu T, 2025. Pro-thrombotic changes in response to ambient ozone exposure exacerbated by temperatures. *Environmental Science & Technology*, 59(17): 8391-8401. https://doi.org/10.1021/acs.est.4c13457. (IF = 11.3)
10. Hua Q, Meng X, Chen W, Xu Y, Xu R, Shi Y, Li J, Meng X, Li A, Chai Q, Sheng M, Yao Y, Fan Y, Qiao R, Zhang Y, Wang T, Zhang Y, Cui X, Yu Y, **Li H**, Tang R, Yan M, Buduo, Danzengdunzhu, Zhuoga, Hou L, Liu Y, Shang J, Chen Q, Qiu X, Ye C, Gong J, Zhu T, 2025. Associations of short-term ozone exposure with hypoxia and arterial stiffness. *Journal of the American College of Cardiology*, 85(6): 606-621. https://doi.org/10.1016/j.jacc.2024.11.044. (IF = 22.3)
11. Gao K, Wang L, Xu Y, Zhang Y, **Li H**, Fu J, Fu J, Lu L, Qiu X, Zhu T, 2024. Concentration identification and endpoint-oriented health risk assessments on a broad-spectrum of organic compounds in atmospheric fine particles: A sampling experimental study in Beijing, China. *Science of the Total Environment*, 906: 167574. https://doi.org/10.1016/j.scitotenv.2023.167574. (IF = 8.0)
12. Tao C, Zhu T, Fu D, Yan B, **Li H**, 2024. Toward better atmospheric polycyclic aromatic hydrocarbons pollution control in the Northern Hemisphere: Process analysis based on interpretable deep learning models. *Journal of Cleaner Production*, 457: 142442. https://doi.org/10.1016/j.jclepro.2024.142442. (IF = 10.0)
13. Qiao R, Chen W, Shi Y, Chai Q, Fan Y, Hua Q, Li A, **Li H**, Li J, Meng X, Sheng M, Xu R, Xu Y, Yao Y, Zhang Y, Zhang Y, Danzengdunzhu, Zhuoga, Zhu T, Gong J, Liu Y, 2024. A comparative analysis on indoor and outdoor $PM_{2.5}$ and their hourly associations with acute respiratory inflammation among college students in Lhasa. *Environmental Science & Technology*, 58: 22668-22677. https://doi.org/10.1021/acs.est.4c04304. (IF = 11.3)
14. Chen W, Han Y, Xu Y, Wang T, Wang Y, Chen X, Qiu X, Li W, **Li H**, Fan Y, Yao Y, Zhu T, 2024. Fine particulate matter exposure and systemic inflammation: A potential mediating role of bioactive lipids. *Science of the Total Environment*, 931: 172993. https://doi.org/10.1016/j.scitotenv.2024.172993. (IF = 8.0)
15. Meng X, Hua Q, Xu R, Shi Y, Zhang Y, Yan M, Chen W, Xu Y, Fan Y, Yao Y, Wang T, Zhang Y, **Li H**, Yu Y, Cui X, Chai Q, Li A, Sheng M, Tang R, Qiao R, Li J, Buduo, Danzengdunzhu, Zhuoga, Hou L, Liu Y, Shang J, Chen Q, Qiu X, Gong J, Zhu T, 2024. A prospective study on the cardiorespiratory effects of air pollution among residents of the Tibetan Plateau. *Hygiene and Environmental Health Advances*, 12: 100115. https://doi.org/10.1016/j.heha.2024.100115. (IF: not yet indexed)
16. **Li H**, Han Y, Wang T, Chen W, Xu Y, Gao K, Wang Y, Gong J, Li W, Zhang H, Wang J, Qiu X, Zhu T, 2023. Development and validation of a simple and rapid HILIC-MS/MS method for the quantification of low-abundant lysoglycerophospholipids in human plasma. *Analytical and Bioanalytical Chemistry*, 415(3): 411-425. https://doi.org/10.1007/s00216-022-04421-9. (IF = 3.8)

17. Wang T, Han Y, Chen X, Chen W, **Li H**, Wang Y, Qiu X, Gong J, Li W, Zhu T, 2023. Particulate air pollution and blood pressure: Signaling by the arachidonate metabolism. *Hypertension*, 80(12): 2687-2696. https://doi.org/10.1161/HYPERTENSIONAHA.123.21410. (IF = 8.2)
18. Xu Y[#], **Li H[#]**, Han Y, Wang T, Wang Y, Gong J, Gao K, Chen W, Li W, Zhang H, Wang J, Qiu X, Zhu T, 2022. A simple and rapid method for extraction and measurement of circulating sphingolipids using LC–MS/MS: A targeted lipidomic analysis. *Analytical and Bioanalytical Chemistry*, 414(6): 2041-2054. https://doi.org/10.1007/s00216-021-03853-z. (**[#]Contributed equally)** (IF = 3.8)
19. Wang T, Chen X, **Li H**, Chen W, Xu Y, Yao Y, Zhang H, Han Y, Zhang L, Que C, Gong J, Qiu X, Zhu T, 2022. Pro-thrombotic changes associated with exposure to ambient ultrafine particles in patients with chronic obstructive pulmonary disease: Roles of lipid peroxidation and systemic inflammation. *Particle and Fibre Toxicology*, 19(1): 65. https://doi.org/10.1186/s12989-022-00503-9. (IF = 8.2)
20. Wang T, Han Y, **Li H**, Fang Y, Liang P, Wang Y, Chen X, Qiu X, Gong J, Li W, Zhu T, 2022. Fine particulate matter and vasoactive 20-hydroxyeicosatetraenoic acid: Insights into the mechanisms of the prohypertensive effects of particulate air pollution. *Science of the Total Environment*, 805: 151298. https://doi.org/10.1016/j.scitotenv.2021.151298. (IF = 8.0)
21. Xu Y, Han Y, Wang Y, Gong J, **Li H**, Wang T, Chen X, Chen W, Fan Y, Qiu X, Wang J, Xue T, Li W, Zhu T, 2022. Ambient air pollution and atherosclerosis: A potential mediating role of sphingolipids. *Arteriosclerosis, Thrombosis, and Vascular Biology*, 42(7): 906-918. https://doi.org/10.1161/ATVBAHA.122.317753. (IF = 7.4)
22. Xu Y, Chen X, Han Y, Chen W, Wang T, Gong J, Fan Y, Zhang H, Zhang L, **Li H**, Wang Q, Yao Y, Xue T, Wang J, Qiu X, Que C, Zheng M, Zhu T, 2022. Ceramide metabolism mediates the impaired glucose homeostasis following short-term black carbon exposure: A targeted lipidomic analysis. *Science of the Total Environment*, 829: 154657. https://doi.org/10.1016/j.scitotenv.2022.154657. (IF = 8.0)
23. Wang T, Han Y, **Li H**, Wang Y, Chen X, Chen W, Qiu X, Gong J, Li W, Zhu T, 2021. Proinflammatory lipid signals trigger the health effects of air pollution in individuals with prediabetes. *Environmental Pollution*, 290: 118008. https://doi.org/10.1016/j.envpol.2021.118008. (IF = 7.3)
24. Wang T, Han Y, **Li H**, Wang Y, Xue T, Chen X, Chen W, Fan Y, Qiu X, Gong J, Xu Y, Wang J, Li W, Zhu T, 2021. Changes in bioactive lipid mediators in response to short-term exposure to ambient air particulate matter: A targeted lipidomic analysis of oxylipin signaling pathways. *Environment International*, 147: 106314. https://doi.org/10.1016/j.envint.2020.106314. (IF = 9.7)
25. Liu J, Chen X, Qiu X, Zhang H, Lu X, **Li H**, Chen W, Zhang L, Que C, Zhu T, 2021. Association between exposure to polycyclic aromatic hydrocarbons and lipid peroxidation in patients with chronic obstructive pulmonary disease. *Science of the Total Environment*, 780: 146660. https://doi.org/10.1016/j.scitotenv.2021.146660. (IF = 8.0)
26. Wang T, **Li H**, Han Y, Wang Y, Gong J, Gao K, Li W, Zhang H, Wang J, Qiu X, Zhu T, 2020. A rapid and high-throughput approach to quantify non-esterified oxylipins for epidemiological studies using online SPE-LC-MS/MS. *Analytical and Bioanalytical Chemistry*, 412(28): 7989-8001. https://doi.org/10.1007/s00216-020-02931-y. (IF = 3.8)

Review Articles (Peer-Reviewed)

27. Paudel D, **Li H**, Holzhausen EA, Young N, Platz EA, Walker DI, Liang D, Aung M, Goodrich JA, Setiawan VW, Le Marchand L, Huang BZ, Conti DV, Chatzi L, Alderete TL, 2025. A scoping review on per- and poly-fluoroalkyl substances (PFAS) and colorectal cancer: Evidence from in vitro, animal, and epidemiological studies. *Environment International*, 203: 109778. https://doi.org/10.1016/j.envint.2025.109778. (IF = 9.7)

Manuscripts Under Revision

28. Holzhausen EA, Jones RB, Young N, **Li H**, Paudel D, Lurmann F, Jones D, Sarnat JA, Chang HH, Goran MI, Liang D, Alderete TL. Prenatal air pollution shapes the longitudinal development of the infant gut microbiome. *Environmental Health Perspectives* (revision submitted).
29. Wang T, Han Y, Wang Z, Chen W, Wang Y, Fan Y, Xu Y, Zhang Y, **Li H**, Liu Y, Zhang H, Yan L, Chatzidiakou L, Gong J, Barratt B, Jones R, Liu J, Wu Y, Kelly F, Zhu T. Winter and summer differences in oxidative changes in response to ambient particles: Roles of sizes and constituents. *Environmental Science & Technology* (revision submitted).
30. **Li H**, Young ND, Paudel D, Holzhausen EA, Brown NR, Ren M, Patel AD, Mullins CE, Bode L, Ottino-González J, Goran MI, Goodrich JA, Conti DV, Eckel SP, Liang D, Walker DI, Chatzi L, Alderete TL. Per- and polyfluoroalkyl substances are associated with metabolic and oligosaccharide composition of human milk across lactation. *Environmental Health Perspectives* (under revision).

Manuscripts Under Review

31. Zhang Y[#], **Li H[#]**, Xu Y, Wang T, Meng X, Hua Q, Xu R, Shi Y, Chen W, Gao Y, Dong Q, Gong J, Zhu T. Signaling lipid responses link short-term ozone exposure to vascular health. *Environmental Science & Technology* (under review). ([#]**Contributed equally**).
32. Marquess KM, **Li H**, Paudel D, Holzhausen EA, Fry Brumit D, Liang D, Hoen AG, Eggers S, Laue HE, Alderete TL. The developing gut microbiome as a link between environmental mixtures and child neurodevelopment. *Current Environmental Health Reports* (under review).
33. Paudel D, Holzhausen EA, **Li H**, Young ND, Jones R, Mullins CE, Patel AD, Tian Y, Anderson BD, Li Z, Bisanz JE, Heinken A, Fässler D, Hertel J, Patterson AD, Goran MI, Goodrich JA, Eckel SP, Jones DP, Walker DI, Liang D, Conti DV, Chatzi L, Alderete TL. Human milk PFAS are associated with disruption of the HMO–Bifidobacterium axis and features of accelerated gut microbiome maturation in a U.S. birth cohort. *Nature Communications* (under review).
34. Cao Y, Han Y, Dong Q, Xu Y, **Li H**, Wang T, Wang Y, Chen X, Chen W, Fan Y, Qiu X, Gong J, Li W, Bu D, Zhu T. Targeted exposome profiling of multi-class emerging contaminants in serum of middle-aged and older adults: Seasonal variability and longitudinal patterns. *Journal of Hazardous Materials* (under review).
35. Wang T, Liao W, Chen X, Que C, Yao Y, Chen W, **Li H**, Xu Y, Zhang L, Gong J, Qiu X, Zhu T. Ferroptosis-related changes in response to short-term ozone exposure. *Environmental Science & Technology* (under review).
36. Zhang K, Zhang Y, Gong J, Chai Q, Li A, Hua Q, Meng X, Chen W, Xu R, Sheng M, Yao Y, Fan Y, **Li H**, Xu Y, Shang J, Liu Y, Zhu T, Qiu X. Associations of $PM_{2.5}$ organic constituents with airway inflammation at low ambient concentrations: A prospective panel study. *Environmental Health Perspectives* (under review).

TEACHING

CLASSROOM INSTRUCTION

Johns Hopkins University

2026 — Guest Speaker. Exposure Sciences and Environmental Epidemiology Graduate Course. "Applications, Opportunities, and Challenges of Microbiome and Metabolomics Research"

Peking University

2019, 2021, 2022 — Environmental Sciences
Teaching assistant, 80–100 undergraduate students

RESEARCH MENTORSHIP

Johns Hopkins University

2026 – Present — Daisy Fry Brumit, PhD, Johns Hopkins University
Provided peer mentorship in epidemiological research methods and exposure assessment

2025 – Present — Katherine M. Marquess, PhD, Johns Hopkins University
Provided peer mentorship in metabolomics-based pesticide exposure analysis and scientific writing

Peking University

2024 – Present — Yi Zhang, PhD candidate, Peking University
Supervised laboratory analyses, data analysis, and manuscript writing for a study on ozone exposure and lipid mediator profiles

2024 – 2025 — Yilin Li, PhD candidate, University of Cambridge
Supervised conceptualization and manuscript writing for a study on the determinants of individual temperature exposure levels

RESEARCH AND PRACTICE FUNDING

Title and Source	*Period*	*Award Total*
NIH NIEHS R01 (R01ES035056) "Pre- and postnatal exposure to air pollutants and the gut microbiome: Implications for brain development in early and mid-childhood". Multiple PI: Alderete (Contact), Peterson; Postdoctoral Research Scholar: **Li**	2024 – 2029	$3,402,514
NIH NIEHS R01 (R01ES035035) "Perfluoroalkyl substances and the gut microbiome and fecal metabolome: Implications for obesity risk in Hispanic children" PI: Alderete; Postdoctoral Research Scholar: **Li**	2023 – 2028	$3,195,629
Maryland Department of Health and Mental Hygiene (Maryland Cigarette Restitution Fund) "Integrating PFAS Exposure, Metabolomics, and Genetics to Address Colorectal Cancer (CRC) Risk" PI: Alderete; Postdoctoral Research Scholar: **Li**	2026	$100,000

PRESENTATIONS

SCIENTIFIC MEETINGS: POSTERS, AND ORALS (FIRST AUTHOR ONLY)

1. **Li H**, Paudel D, Marquess KM, Jia Q, Ottino-González J, Lurmann FW, Conti DV, Walker DI, Chang HH, Liang D, Bansal R, Goran MI, Peterson BS, Alderete TL, 2026. Longitudinal gut microbiome and fecal metabolome signatures linking prenatal fine particulate matter component mixture exposure and child neurodevelopment. *International Society for Environmental Epidemiology (**ISEE**) Annual Conference, Munich, Germany*. *(Oral presentation)*
2. **Li H**, Schenker RB, Marquess KM, Paudel D, Ramirez CB, Lurmann FW, Ren M, Jang C, Allayee H, Liao J, Chen Z, Goran MI, Alderete TL, 2026. Ambient fine particulate matter components and liver fat and stiffness in adolescents: Metabolic mechanisms and PNPLA3 genotype-specific associations. *Health Effects Institute (**HEI**) Annual Conference, Chicago, IL, USA*; also presented at the *International Society for Environmental Epidemiology – North American Chapter (**ISEE-NAC**) Conference, Baltimore, MD, USA*. *(Poster presentation)*
3. **Li H**, Holzhausen EA, Young ND, Paudel D, Ottino-González J, Lurmann FW, Walker DI, Chang HH, Liang D, Bansal R, Goran MI, Peterson BS, Alderete TL, 2025. Associations of fine particulate matter components with neurodevelopment in early childhood: Evidence from mixture approaches. *International Society for Environmental Epidemiology (**ISEE**) Annual Conference, Atlanta, GA, USA*. *(Oral presentation)*

INVITED SEMINARS AND WORKSHOPS

National

4. **Li H**, 2026. "Early-life health effects of PFAS in human milk: Impacts on milk composition, the infant gut microbiome, and neurodevelopment." *The Leslie Isaacs Special Lecture, Johns Hopkins Medicine Sherrilyn and Ken Fisher Center for Environmental Infectious Diseases, Baltimore, MD*.

International

5. Alderete TL, **Li H**, 2026. "Early-life health effects of PFAS in human milk: Impacts on milk composition, the infant gut microbiome, and neurodevelopment." *Boya International Forum on Environmental Research, Department of Environmental Health, College of Environmental Sciences and Engineering, Peking University, Beijing, China*.

TECHNICAL SKILLS

FIELDWORK

Design and implementation of environmental epidemiology field studies; operation and maintenance of air pollution monitoring instruments; cardiopulmonary health assessments

LABORATORY

Sample preparation for metabolomics and transcriptomics analyses; liquid chromatography–mass spectrometry (LC–MS) using triple quadrupole (QqQ) and quadrupole time-of-flight (QToF) platforms; flow cytometry; ultracentrifugation; PCR and gel electrophoresis

DATA ANALYSIS

Linear, non-linear, and mixed-effects modeling; environmental mixture analysis; mediation and high-dimensional mediation analysis; metabolomics data preprocessing; metabolomics pathway enrichment analysis; microbiome analysis; multi-omics integration; and scientific visualization using R, BioRender, and Adobe Illustrator